Physics 9702 Paper 4 · Topic 19

Capacitance

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/42 February/March 2023, Question 5 (10 marks) page 2

Examiner report: Only the strongest candidates were able to correctly determine the resistance of R. (d) Most candidates realised that a straight line from the same initial value of ln Q was required, but only the strongest candidates correctly deduced the quantitative effect of the parallel resistor on the gradient of the line.

2 9702/41 October/November 2021, Question 6 (5 marks) page 4

Examiner report: The expression for the energy stored was more challenging. (b) These expressions proved to be challenging for candidates.

3 9702/42 February/March 2022, Question 5 (10 marks) page 6

Examiner report: (ii) Only the very strongest candidates answered this correctly, with most candidates forgetting that energy is proportional to V2 and attempting to apply the general exponential decay equation incorrectly using x as the energy stored.

4 9702/42 October/November 2021, Question 6 (7 marks) page 9

Examiner report: (c) Most candidates found this to be a challenging question. Only the strongest candidates appreciated that conservation of charge means that the total charge remains the same when distributed across the two spheres.

5 9702/42 October/November 2022, Question 6 (10 marks) page 10

Examiner report: (a) Candidates found it difficult to demonstrate understanding of why the discharge of a capacitor through a resistor is exponential. Only the strongest candidates were able to give a complete explanation.

6 9702/41 May/June 2021, Question 7 (9 marks) page 12

Examiner report: A large number of power-of-ten errors were made as candidates found it difficult to convert their value into pF. (c) This was another challenging explanation.

7 9702/42 May/June 2023, Question 5 (9 marks) page 14

Examiner report: (b) (i) Many candidates found this question difficult. (ii) Many candidates found it difficult to give a coherent answer to this question, with responses often containing unexplained algebra.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

5 A capacitor, a battery of electromotive force (e.m.f.) 12 V, a resistor R and a two-way switch are connected in the circuit shown in Fig. 5.1.

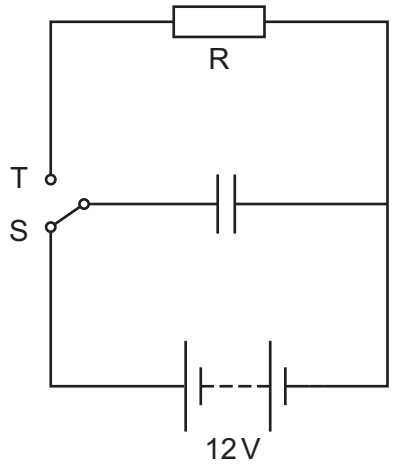


Fig. 5.1

The switch is initially in position S. When the capacitor is fully charged, the switch is moved to position T so that the capacitor discharges. At time t after the switch is moved the charge on the capacitor is Q.

The variation with t of $\ln(Q/\mu C)$ is shown in Fig. 5.2.

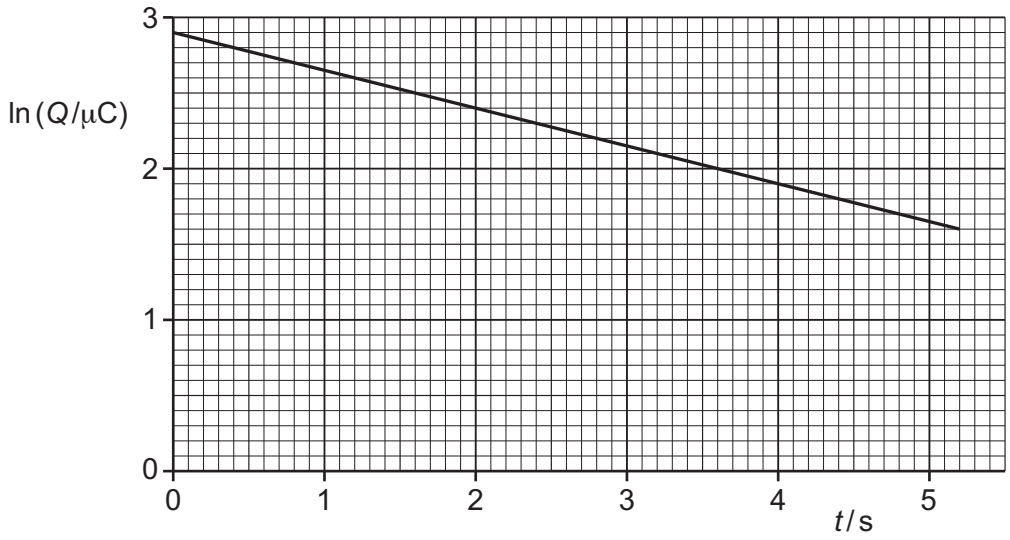


Fig. 5.2

(a) Show that the capacitance of the capacitor is 1.5 μF.

[3]

(b) Determine the resistance of R.

resistance = ... Ω [3]

(c) Calculate the energy stored in the capacitor at time $t = 0$.

energy = ... J [2]

(d) A second identical resistor is now connected in parallel with R.

The switch is initially in position S. When the capacitor is fully charged, the switch is moved to position T so that the capacitor discharges. At time t after the switch is moved the charge on the capacitor is Q.

On Fig. 5.2, sketch a line to show the variation of $\ln(Q/\mu C)$ with t between time $t = 0$ and time $t = 5.0\,s$. [2]

[Total: 10]

6 **(a)** A capacitor consists of two parallel metal plates, separated by air, at a variable distance x apart, as shown in Fig. 6.1. The capacitance C is inversely proportional to x.

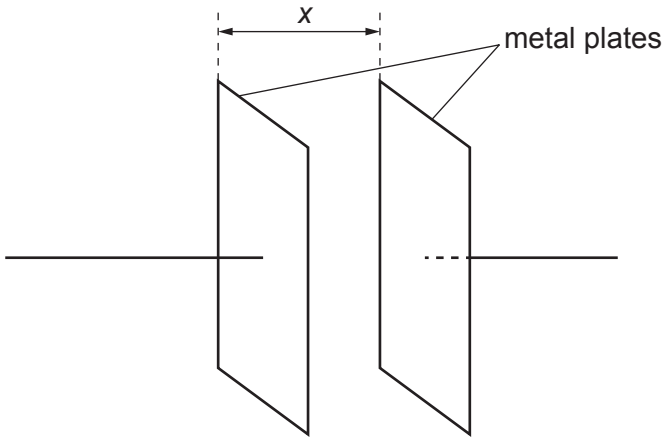


Fig. 6.1

The capacitor is charged by a supply so that there is a potential difference (p.d.) V between the plates.

State expressions, in terms of C and V, for the charge Q on one of the plates and for the energy E stored in the capacitor.

Q = ... E = ... [1]

(b) The charged capacitor in **(a)** is now disconnected from the supply. The plates of the capacitor are initially separated by distance L. They are then moved closer together by a distance D, as shown in Fig. 6.2.

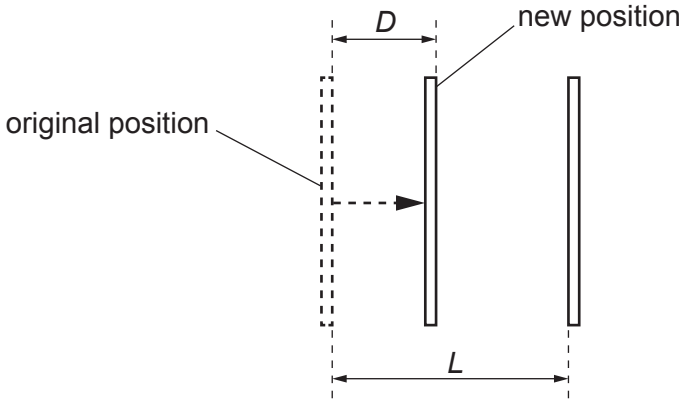


Fig. 6.2

State expressions, in terms of C, V, L and D, for:

(i) the new capacitance C_N

C_N = ... [1]

(ii) the new charge Q_N on one of the plates

Q_N = ... [1]

(iii) the new p.d. V_N between the plates.

V_N = ... [1]

(c) Explain whether reducing the separation of the plates in **(b)** results in an increase or decrease in the energy stored in the capacitor.

.....

.....

..... [1]

[Total: 5]

5 The variation with potential difference V of the charge Q on one of the plates of a capacitor is shown in Fig. 5.1.

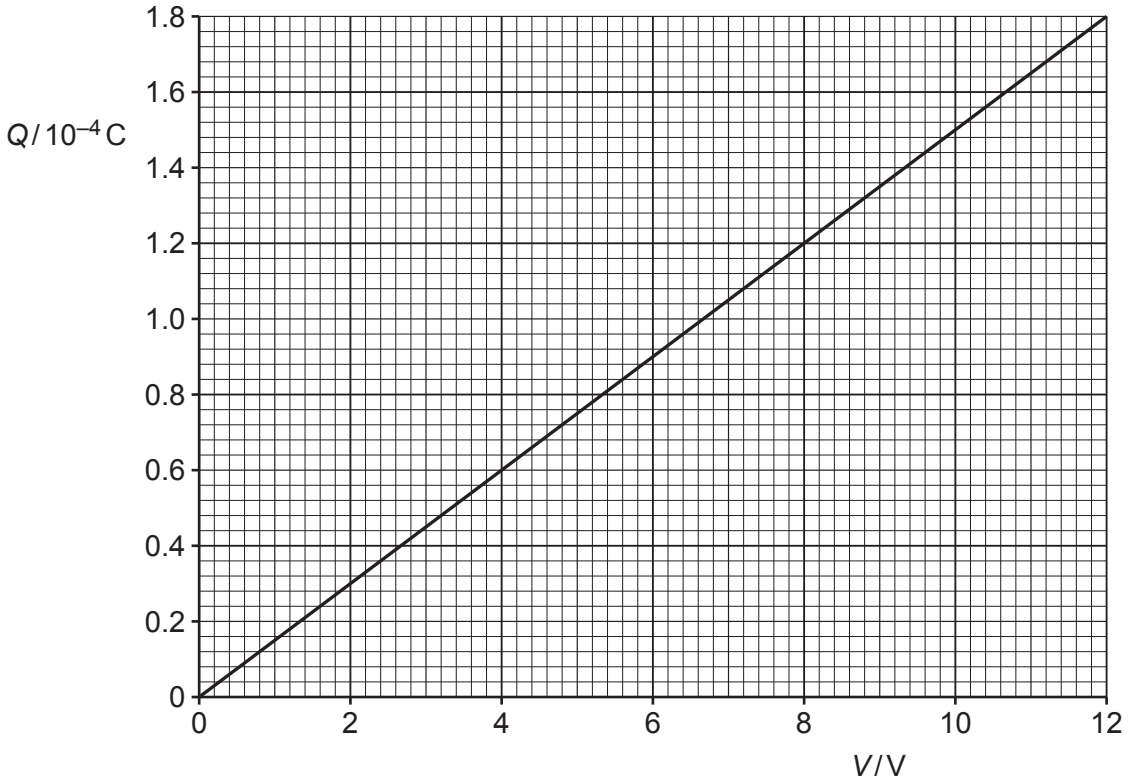


Fig. 5.1

The capacitor is connected to an 8.0 V power supply and two resistors R and S as shown in Fig. 5.2.

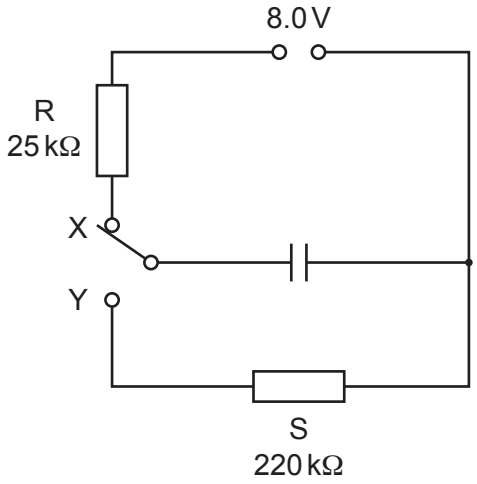


Fig. 5.2

The resistance of R is 25 kΩ and the resistance of S is 220 kΩ.

The switch can be in either position X or position Y.

(a) The switch is in position X so that the capacitor is fully charged.

Calculate the energy E stored in the capacitor.

E = ... J [2]

(b) The switch is now moved to position Y.

(i) Show that the time constant of the discharge circuit is 3.3 s.

[2]

(ii) The fully charged capacitor in **(a)** stores energy E.

Determine the time t taken for the stored energy to decrease from E to $E/9$.

t = ... s [4]

(c) A second identical capacitor is connected in parallel with the first capacitor.

State and explain the change, if any, to the time constant of the discharge circuit.

.....

.....

..... [2]

[Total: 10]

6 **(a)** Define *electric potential*.

.....

.....

..... [2]

(b) An isolated conducting sphere in a vacuum has radius r and is initially uncharged. It is then charged by friction so that it carries a final charge Q. This charge can be considered to be acting at the centre of the sphere.

By considering the electric potential at its surface, show that the capacitance C of the sphere is given by

$$C = 4\pi\varepsilon_0 r$$

where ε_0 is the permittivity of free space.

[2]

(c) The dome of an electrostatic generator is a spherical conductor of radius 13 cm. It is initially charged so that the electric potential at the surface is 4.5 kV.

A smaller isolated sphere of radius 5.2 cm, initially uncharged, is brought near to the dome. Sparking causes a current between the two spheres until they reach the same potential. Assume that any charge on a sphere may be considered to act as a point charge at its centre.

Calculate the charge that is transferred between the two spheres.

charge = ... C [3]

[Total: 7]

6 A capacitor of capacitance C and a resistor of resistance R are connected as shown in Fig. 6.1.

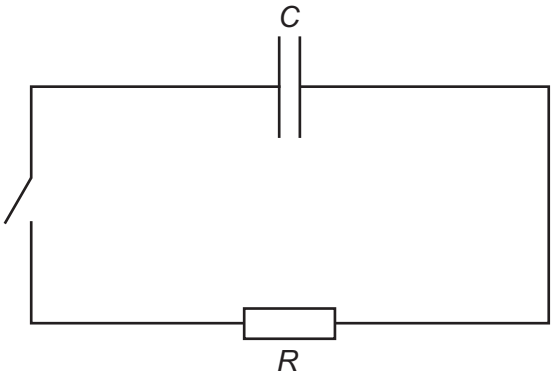


Fig. 6.1

Initially, the capacitor is charged and the switch is open.

The switch is closed at time $t = 0$.

Fig. 6.2 and Fig. 6.3 show, respectively, the variations with t of the charge Q on the capacitor and the potential difference (p.d.) V across the resistor.

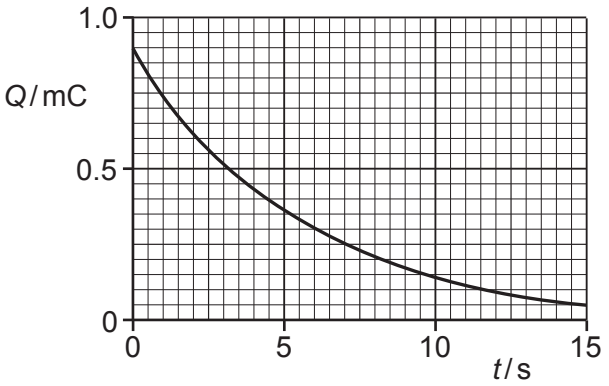


Fig. 6.2

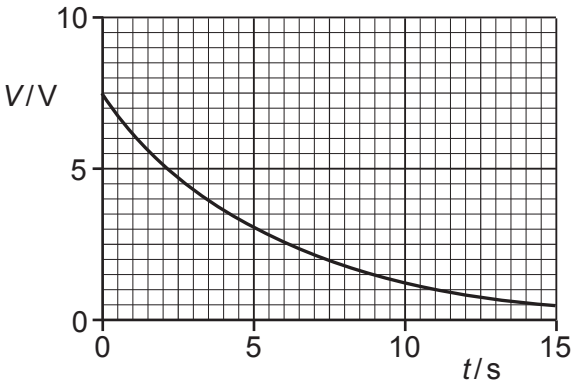


Fig. 6.3

(a) Explain the shape of the line in Fig. 6.3 representing the variation of V with t.

.....

.....

.....

.....

..... [3]

(b) Use Fig. 6.2 to show that the time constant of the circuit in Fig. 6.1 is 5.5 s.

[3]

(c) Use Fig. 6.2, Fig. 6.3 and the information in **(b)** to determine:

(i) capacitance C, in μF

C = ... μF [2]

(ii) resistance R, in kΩ.

R = ... kΩ [2]

[Total: 10]

7 **(a)** State what is meant by the *capacitance* of a parallel plate capacitor.

.....

.....

..... [2]

(b) A capacitor of capacitance C is connected into the circuit shown in Fig. 7.1.

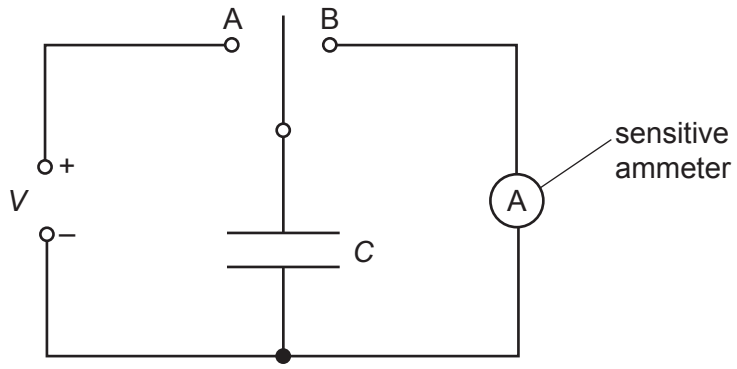


Fig. 7.1

When the two-way switch is in position A, the capacitor is charged so that the potential difference across it is V.
The switch moves to position B and the capacitor fully discharges through the sensitive ammeter.

The switch moves repeatedly between A and B so that the capacitor charges and then discharges with frequency f.

(i) Show that the average current I in the ammeter is given by the expression

$$I = fCV.$$

[2]

(ii) For a potential difference V of 150 V and a frequency f of 60 Hz, the average current in the ammeter is 4.8 μA.

Calculate the capacitance, in pF, of the capacitor.

capacitance = ... pF [2]

(c) A second capacitor, having the same capacitance as the capacitor in **(b)**, is connected into the circuit of Fig. 7.1. The two capacitors are connected in series.

State and explain the new reading on the ammeter.

new reading = ... μA

.....

.....

..... [3]

[Total: 9]

5 Two capacitors A and B are connected into the circuit shown in Fig. 5.1.

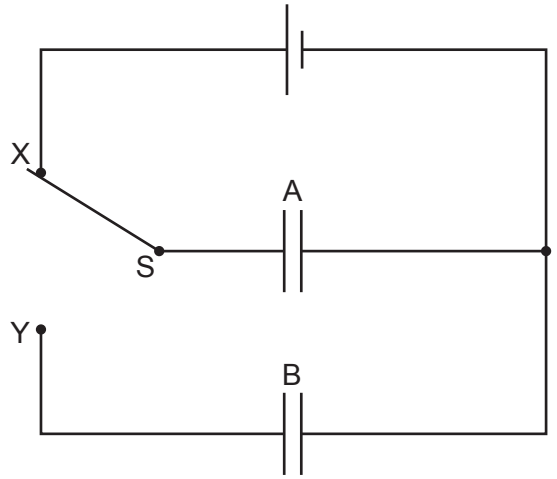


Fig. 5.1

Capacitor A has capacitance C and capacitor B has capacitance $3C$.
The electromotive force (e.m.f.) of the cell is V.
The two-way switch S is initially at position X, and capacitor B is initially uncharged.

(a) State, in terms of V and C, expressions for:

(i) the initial charge Q_A on the plates of capacitor A

Q_A = ... [1]

(ii) the initial energy E_A stored in capacitor A.

E_A = ... [1]

(b) The two-way switch S is now moved to position Y.

(i) State and explain what happens to the charge that was initially on the plates of capacitor A.

.....

.....

..... [2]

(ii) Show that the final potential difference (p.d.) V_B across capacitor B is given by

$$V_B = \frac{V}{4}.$$

Explain your reasoning.

[3]

(iii) Determine an expression, in terms of V and C, for the decrease ΔE in the total energy that is stored in the capacitors as a result of the change of the position of the switch.

ΔE = ... [2]

[Total: 9]

Question	Answer	Marks
5(a)	from graph ln Q = 2.9 (so Q = 18.2 μC)	**B1**
	$C = Q / V$	**C1**
	= 18.2 / 12 = 1.5 μF	**A1**
5(b)	gradient = –0.25	**C1**
	gradient = $-1 / RC$	**C1**
	$R = 1 / (0.25 \times 1.5 \times 10^{-6})$ $= 2.7 \times 10^{6}\ \Omega$	**A1**
	or $\frac{Q}{Q_0} = e^{-t/CR}$ **or** ln Q – ln $Q_0 = \frac{-t}{CR}$	**(C1)**
	e.g. $\frac{4.95}{18.2} = e^{-5.2/(1.5\times10^{-6}R)}$ **or** 1.6 – 2.9 = 5.2 / ($1.5 \times 10^{-6}R$)	**(C1)**
	$R = 2.7 \times 10^{6}\ \Omega$	**(A1)**

Question	Answer	Marks
5(c)	$W = ½ QV$	**C1**
	$= ½ \times 18.2 \times 10^{-6} \times 12$ $= 1.1 \times 10^{-}4$ J	**A1**
	or $W = ½ CV^2$	**(C1)**
	$= ½ \times 1.5 \times 10^{-6} \times 12^2$ $= 1.1 \times 10^{-}4$ J	**(A1)**
	or $W = ½ Q^2 / C$	**(C1)**
	$= ½ \times (18.2 \times 10^{-6})^2 / 1.5 \times 10^{-6}$ $= 1.1 \times 10^{-}4$ J	**(A1)**
5(d)	straight line with different negative gradient starting from (0, 2.9)	**M1**
	straight line between $t = 0$ and at least $t = 5.0$ s with twice the gradient of the original line	**A1**

Question	Answer	Marks
6(a)	$Q = CV$ **and** $E = ½CV^2$	**B1**
6(b)(i)	$C_N = CL / (L – D)$	**B1**
6(b)(ii)	(charge is unchanged by moving the plates so) $Q_N = CV$	**B1**
6(b)(iii)	$V_N = Q_N / C_N$ $= (CV) / [CL / (L – D)]$ $= V(L – D) / L$	**B1**
6(c)	oppositely charged plates attract, so energy stored decreases	**B1**

Question	Answer	Marks
5(a)	(energy stored =) area under line or ½ QV = ½ × 8.0 × 1.2 × 10^{-4}	**C1**
	= **4.8 × 10^{-4}** J	**A1**
5(b)(i)	(τ=) RC	**C1**
	(τ=) 220 × 10^{3} × (1.2 × 10^{-4}/8.0) = **3.3** s	**A1**
5(b)(ii)	$E \propto V^2$	**C1**
	(so time to) V_o / 3 $V = V_o e^{-t/RC}$	**C1**
	$\frac{V_o}{3} = V_o e^{-t/3.3}$ $\frac{1}{3} = e^{-t/3.3}$	**C1**
	t = 3.6 s	**A1**
5(c)	(total) capacitance is doubled	**M1**
	time constant is doubled	**A1**

Question	Answer	Marks
6(a)	work done per unit charge	**B1**
	(work done in) moving positive charge from infinity	**B1**
6(b)	$C = Q / V$	**C1**
	$V = Q / (4\pi\varepsilon_0 r)$ and so $C = Q / [Q / (4\pi\varepsilon_0 r)] = 4\pi\varepsilon_0 r$	**A1**
6(c)	$Q = 4\pi\varepsilon_0 rV \quad = 4\pi \times 8.85 \times 10^{-12} \times 0.13 \times 4500$ $(= 6.5 \times 10^{-8}\ \text{C})$	**C1**
	$(Q - q) / 13 = q / 5.2$	**C1**
	$5.2Q - 5.2q = 13q$, so $q = (5.2 / 18.2)Q$ $q \quad = (5.2 / 18.2) \times 6.5 \times 10^{-8}$ $= 1.9 \times 10^{-8}\ \text{C}$	**A1**
	or	
	$V_T = Q_T / C_T$ $= 6.5 \times 10^{-8} / [4\pi \times 8.85 \times 10^{-12} \times (0.13 + 0.052)]$ $(= 3210\ \text{V})$	**(C1)**
	$q \quad = 4\pi \times 8.85 \times 10^{-12} \times 0.052 \times 3210$ $= 1.9 \times 10^{-8}\ \text{C}$	**(A1)**

Question	Answer	Marks
6(a)	• p.d. across resistor = p.d. across capacitor • current (in resistor) proportional to p.d. across it • current causes capacitor to lose charge • charge (on capacitor) proportional to p.d. so p.d. decreases *Any two points, 1 mark each*	**B2**
	rate of change of p.d. decreases as p.d. decreases	**B1**
6(b)	Q_0 = 0.90 mC **and** at t = one time constant, $Q = Q_0$ exp (–1)	**B1**
	at t = one time constant, Q = 0.90 exp (–1) = 0.33 mC	**M1**
	evidence of graph reading: when Q = 0.33 mC, t = 5.5 s	**A1**
	or	
	evidence of two correct sets of readings for Q and t from the graph	**(B1)**
	correct substitution of Q and t values into $Q_2 = Q_1$ exp $[(t_1 - t_2) / \tau]$	**(M1)**
	calculation to give τ = 5.5 s	**(A1)**
	or	
	read-off of half-life as 3.75 s	**(B1)**
	use of $Q = Q_0$ exp $(-t / \tau)$ to show that τ = half-life / ln 2	**(M1)**
	τ = 3.75 / ln 2 = 5.4 s	**(A1)**

Question	Answer	Marks
6(b)	**or**	
	tangent drawn on Q–t graph and value of Q at exact same time as tangent read from graph	**(M1)**
	gradient of tangent correctly calculated	**(A1)**
	$\tau = Q$ / gradient used to correctly calculate a value for τ as 5.5 s	**(A1)**
6(c)(i)	$C = Q / V$	**C1**
	$= [(0.90 \times 10^{-3}) / 7.5] = 1.2 \times 10^{-4}$ C $= 120\ \mu$F	**A1**
6(c)(ii)	$R = \tau / C$	**C1**
	$= 5.5 / (1.2 \times 10^{-4})$ (= 45 800 Ω) = 46 kΩ	**A1**

Question	Answer	Marks
7(a)	charge / potential	**M1**
	charge is on one plate, potential is p.d. between the plates	**A1**
7(b)(i)	$I = Q / t$	**M1**
	charge = CV and time = $1 / f$ leading to $I = fCV$	**A1**
7(b)(ii)	$4.8 \times 10^{-6} = 150 \times 60 \times C$	**C1**
	C = 530 pF	**A1**
7(c)	(total) capacitance is halved	**B1**
	charge (for each cycle/discharge) is halved **or** since f and V are constant, current is proportional to capacitance	**B1**
	current = 2.4 μA	**B1**

Question	Answer	Marks
5(a)(i)	$Q_A = CV$	**A1**
5(a)(ii)	$E_A = ½CV^2$	**A1**
5(b)(i)	some of the charge transfers to (the plates of) capacitor B	**B1**
	transfer is because the p.d.s across the capacitors are not equal **or** transfer stops when the p.d.s across the capacitors become equal	**B1**
5(b)(ii)	$V_A = V_B$	**M1**
	charge on A + charge on B = CV	**M1**
	$CV_B + 3CV_B = CV$ leading to $V_B = V / 4$	**A1**
	or	
	$C_T = 4C$	**(M1)**
	$Q_T = CV$	**(M1)**
	$V_B = CV / 4C = V / 4$	**(A1)**
5(b)(iii)	$\Delta E = ½CV^2 - nCV^2$, where n is a multiple that is less than ½ **or** total final energy $= ½ \times 4C \times (V / 4)^2$ $= ⅛CV^2$	**C1**
	$\Delta E = ½CV^2 - ⅛CV^2$ $= ⅜CV^2$	**A1**